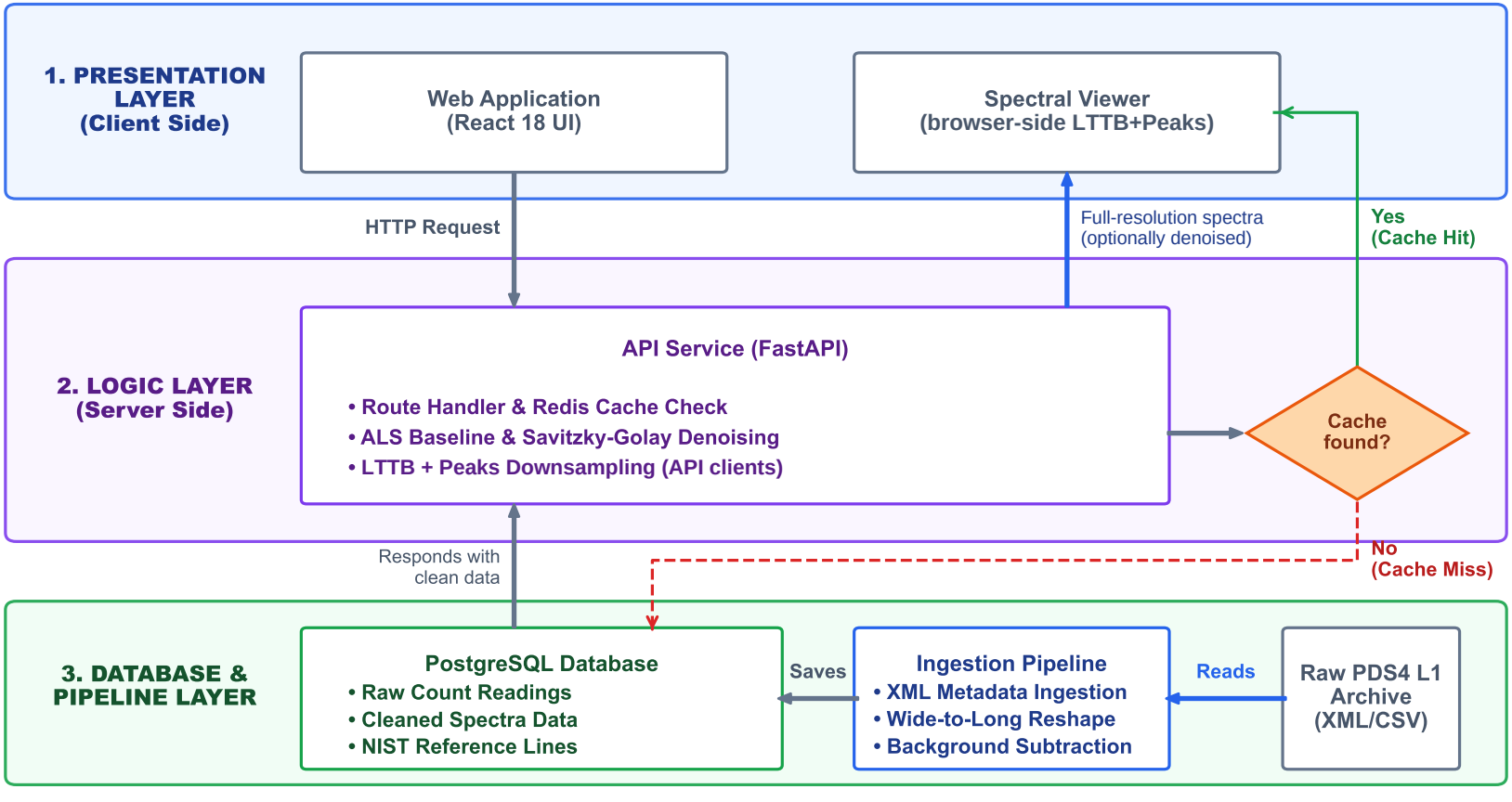
1. PRESENTATION LAYER (Client Side)
Web Application (React 18 UI)
Spectral Viewer (browser-side LTTB+Peaks)
HTTP Request
Full-resolution spectra (optionally denoised)
Yes (Cache Hit)
2. LOGIC LAYER (Server Side)
API Service (FastAPI)
• Route Handler & Redis Cache Check
• ALS Baseline & Savitzky-Golay Denoising
• LTTB + Peaks Downsampling (API clients)
Cache found?
No (Cache Miss)
Responds with clean data
3. DATABASE & PIPELINE LAYER
PostgreSQL Database
• Raw Count Readings
• Cleaned Spectra Data
• NIST Reference Lines
Saves
Ingestion Pipeline
• XML Metadata Ingestion
• Wide-to-Long Reshape
• Background Subtraction
Reads
Raw PDS4 L1 Archive (XML/CSV)